

★ UDEMY BESTSELLER

DATA SCIENCE & AI

OFFICIAL COMPLETE SYLLABUS

UPDATED 2026-08-01

Data Science & AI Masters 2026 - From Python To Gen AI

Master Data Science and AI: Learn Python, EDA, Stats, SQL, Machine Learning, NLP, Deep Learning and Gen AI

LEAD INSTRUCTOR

**Dr. Satyajit
Pattnaik**

TOTAL DURATION

**115h 56m (25
Sections)**

TOTAL CURRICULUM
ITEMS

**632 Lectures +
12 Quizzes**

STUDENT
COMMUNITY

**★ 4.4 (2,689
reviews)**

Key Competencies & What You'll Learn

✓ Build a solid foundation in Python programming to effectively implement AI concepts and applications.

✓ Learn how Machine Learning & Deep Learning works

✓ Learn how transformer models revolutionize NLP tasks, and how to leverage them for various applications.

✓ Gain hands-on experience with Retrieval-Augmented Generation (RAG) and Langchain for building advanced AI applications.

✓ Learn how to utilize vector databases for efficient storage and retrieval of embeddings in AI projects.

✓ Understand the complete pipeline of Natural Language Processing, from data preprocessing to model deployment.

✓ Explore the essentials of Large Language Models (LLMs) and their applications in generative tasks.

✓ Develop skills in crafting effective prompts to optimize model performance and achieve desired outputs.

Target Audience

- Complete beginners willing to jump into the field of Data Science, AI & Gen AI
- Experienced Professionals willing to switch to Data Science, AI & Gen AI

Prerequisites & Requirements

- No programming experience is needed.
- A laptop with atleast 8gb memory is recommended (4gb memory is also fine, can use Google Colab as a backup)
- No prior knowledge on SQL, or Machine Learning is needed

Curriculum Roadmap & Master Table of Contents

SEC #	MODULE / SECTION TITLE	LECTURES	QUIZZES	RESOURCES	DURATION
Phase 1: Foundations of Data Science & Programming — Core Python mastery, mathematical statistics, exploratory data analysis, and relational databases.					
01	Introduction	3	-	-	4m
02	Python for Data Science	22	1	2	6h 35m
03	Business Statistics	58	1	-	5h 25m
04	Exploratory Data Analysis	35	1	2	5h 05m
05	SQL for Data Science	19	1	1	5h 10m
Phase 2: Machine Learning & Time Series Forecasting — Supervised & unsupervised ML, loss functions, optimization, and time series modeling with forecasting.					
06	Machine Learning	86	1	1	20h 13m
07	Time Series Analysis & Forecasting	60	1	1	7h 29m
Phase 3: Deep Learning, Neural Networks & NLP — ANN, CNN, RNN architectures, word embeddings, sequence models, and classical NLP pipelines.					
08	Deep Learning & Neural Networks	34	1	1	5h 57m
09	Natural Language Processing	47	1	3	9h 21m
Phase 4: Generative AI, Transformers & LLMs — Attention mechanisms, transformers, Hugging Face, LLMs, prompt engineering, fine-tuning, and RAG evaluation.					
10	Transformers & Generative AI	53	1	9	11h 12m
11	Generative AI (Advanced)	9	1	-	1h 47m
12	RAG Assessment and Evaluation Metrics	12	1	3	3h 45m
Phase 5: Agentic AI & Autonomous Multi-Agent Systems — AI agent architectures, tool use, LangGraph stateful multi-agent workflows, and no-code automation with n8n.					
13	Introduction to Agentic AI	19	-	5	3h 24m
14	Agentic AI using LangGraph	13	-	1	1h 33m
15	No-Code Tools for Agentic AI	9	-	3	2h 02m

SEC #	MODULE / SECTION TITLE	LECTURES	QUIZZES	RESOURCES	DURATION
Phase 6: Production Engineering & Cloud Deployment — Model serving, REST APIs, Docker containerization, AWS EC2 deployments, and data engineering pipelines.					
16	ML/DL Deployment	7	-	1	1h 49m
17	Data Engineering Basics	11	1	-	39m
23	Generative AI Deployment	9	-	1	58m
Phase 7: End-to-End Industry Capstone Projects — Real-world portfolio projects spanning healthcare, solar energy, wildlife conservation, GenAI and AI agents.					
18	Machine Learning Project #1 - Cancer Risk Prediction	22	-	3	2h 53m
19	Deep Learning Project #1 - Solar Panel Defect Classification	18	-	2	2h 34m
20	Deep Learning Project #2 - Elephant Species Classification	16	-	3	2h 38m
21	Generative AI Projects	32	-	10	8h 56m
22	Agentic AI Projects	7	-	1	1h 24m
Phase 8: Career Mastery & Client Role-Play — Comprehensive interview preparation questions and simulated client consulting scenarios.					
24	Interview Prep (NEW SECTION)	31	-	1	4h 51m
25	Real-Time Client Discussions (Role Play)	0	4 roles	-	0m
TOTAL	25 Modules / Complete Curriculum	632	12 + 4 roles	54	115h 56m

Complete Detailed Syllabus (Module by Module)

SECTION 01 Introduction 3 Lectures 4m				
#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Welcome Page	VIDEO	FREE PREVIEW	02:23
2	Introduction to the Course	ARTICLE	—	01:54
3	Course Resources	ARTICLE	—	00:21

SECTION 02 Python for Data Science

22 Lectures

1 Quiz

2 Files

6h 35m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Welcome to the module	ARTICLE	 Resources	01:31
2	Let's install Python together!!	VIDEO	 Python Installation Guide.pdf	17:38
3	Google Colab, what's that?	VIDEO	—	03:47
4	Let's leverage chatGPT for help!!	VIDEO	—	05:11
5	Introduction to Python	VIDEO	—	04:57
6	Variables & Keywords	VIDEO	—	19:54
7	Datatypes Operators	VIDEO	—	21:49
8	Lists	VIDEO	—	27:51
9	Tuples	VIDEO	—	24:57
10	Sets	VIDEO	—	12:45
11	Dictionary	VIDEO	—	17:39
12	Loops & Iterations	VIDEO	—	19:16
13	Functions	VIDEO	—	23:31
14	Map Reduce Filter	VIDEO	—	30:15
15	File Handling	VIDEO	—	19:19
16	Control Structures	VIDEO	—	09:39
17	OOPs	VIDEO	—	20:42
18	NumPy	VIDEO	—	28:33
19	Pandas	VIDEO	FREE PREVIEW	34:13
20	Data Visualization	VIDEO	—	06:12
21	Matplotlib	VIDEO	—	26:37
22	Seaborn	VIDEO	—	19:10
23	Python Quiz	QUIZ	—	Assessment

SECTION 03 Business Statistics

58 Lectures

1 Quiz

5h 25m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Course Contents	ARTICLE	—	00:03
2	Introduction	VIDEO	FREE PREVIEW	05:50
3	Types of Data (Agenda)	VIDEO	FREE PREVIEW	00:38
4	Descriptive Stats	VIDEO	—	08:10
5	Inferential Stats	VIDEO	—	02:41
6	Qualitative Data	VIDEO	—	08:33
7	Quantitative Data	VIDEO	—	04:17
8	Sampling Techniques (Agenda)	VIDEO	—	01:37
9	Population vs Sample	VIDEO	—	03:47
10	Why Sampling is important	VIDEO	—	03:20
11	Types of Sampling	VIDEO	—	03:32
12	Cluster Random Sampling	VIDEO	—	05:12
13	Probability Sampling	VIDEO	—	08:08
14	Non probability sampling	VIDEO	—	05:18
15	Population Sampling	VIDEO	—	08:11
16	Why n-1 and not n	VIDEO	—	05:06
17	Descriptive Analytics (Agenda)	VIDEO	—	01:06
18	Measures of Central Tendency	VIDEO	—	02:30
19	Mean	VIDEO	—	05:31
20	Median	VIDEO	—	07:07
21	Mode	VIDEO	—	04:39
22	Measures of Dispersion	VIDEO	—	04:21
23	Range	VIDEO	—	01:43
24	IQR	VIDEO	—	04:49
25	Variance Standard Deviation	VIDEO	—	09:56
26	Mean Deviation	VIDEO	—	04:12

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
27	Probability (Agenda)	VIDEO	—	00:59
28	Probability	VIDEO	—	07:09
29	Addition Rule	VIDEO	—	07:17
30	Independent Events	VIDEO	—	04:39
31	Cumulative Probability	VIDEO	—	04:26
32	Conditional Probability	VIDEO	—	07:57
33	Bayes Theorem 1	VIDEO	—	02:08
34	Bayes Theorem 2	VIDEO	—	04:59
35	Probability Distrubution (Agenda)	VIDEO	—	02:04
36	Uniform Distribution	VIDEO	—	08:02
37	Binomial Distribution	VIDEO	—	12:48
38	Poisson Distribution	VIDEO	—	03:29
39	Normal Distribution Part 1	VIDEO	—	10:34
40	Normal Distribution Part 2	VIDEO	—	07:38
41	Skewness	VIDEO	—	04:50
42	Kurtosis	VIDEO	—	02:36
43	Calculating Probability with Z-score for Normal Distribution Part 1	VIDEO	—	08:12
44	Calculating Probability with Z-score for Normal Distribution Part 2	VIDEO	—	04:45
45	Calculating Probability with Z-score for Normal Distribution Part 3	VIDEO	—	03:33
46	Covariance & Correlation (Agenda)	VIDEO	—	00:31
47	Covariance	VIDEO	—	12:06
48	Correlation	VIDEO	—	16:06
49	Covariance VS Correlation	VIDEO	—	03:57
50	Hypothesis Testing	VIDEO	—	06:59
51	Tailed Tests	VIDEO	—	02:37
52	p-value	VIDEO	—	04:14

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
53	Types of Test	VIDEO	—	08:22
54	T Test	VIDEO	—	07:32
55	Z Test	VIDEO	—	09:24
56	Chi Square Test	VIDEO	—	08:58
57	ANOVA	VIDEO	—	08:56
58	Correlation Test (Practicals)	VIDEO	—	06:57
59	Practice Quiz	QUIZ	—	Assessment

SECTION 04 Exploratory Data Analysis

35 Lectures

1 Quiz

2 Files

5h 05m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Course Contents	ARTICLE	 Resources	00:08
2	DA,DS Processes	VIDEO	—	06:35
3	What is EDA	VIDEO	—	03:08
4	Visualization	VIDEO	—	04:49
5	Agenda	VIDEO	—	02:32
6	Steps involved in EDA (Data Sourcing)	VIDEO	—	04:35
7	Steps involved in EDA (Data Cleaning)	VIDEO	—	04:11
8	Handle Missing Values (Theory)	VIDEO	—	06:25
9	Handle Missing Values (Practicals)	VIDEO	—	11:35
10	Feature Scaling (Theory)	VIDEO	—	11:04
11	Standardization Example	VIDEO	—	04:13
12	Normalization Example	VIDEO	—	02:30
13	Feature Scaling (Practicals)	VIDEO	—	13:50
14	Outlier Treatment (Theory)	VIDEO	—	09:21
15	Outlier Treatment (Practicals)	VIDEO	—	13:41
16	Invalid Data	VIDEO	—	04:47
17	Types of Data	VIDEO	—	02:57
18	Types of Analysis	VIDEO	—	02:35
19	Univariate Analysis	VIDEO	—	09:00
20	Bivariate Analysis	VIDEO	—	05:15
21	Multivariate Analysis	VIDEO	—	01:21
22	Numerical Analysis	VIDEO	—	03:56
23	Analysis Practicals	VIDEO	—	30:14
24	Derived Metrics	VIDEO	—	04:33
25	Feature Binning (Theory)	VIDEO	—	07:17
26	Feature Binning (Practicals)	VIDEO	—	10:48

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
27	Feature Encoding (Theory)	VIDEO	—	12:25
28	Feature Encoding (Practicals)	VIDEO	—	21:32
29	Case Study	VIDEO	—	18:15
30	Data Exploration	VIDEO	—	19:16
31	Data Cleaning	VIDEO	—	09:45
32	Univariate Analysis	VIDEO	—	14:17
33	Bivariate Analysis Part 1	VIDEO	—	14:59
34	Bivariate Analysis Part 2	VIDEO	—	08:50
35	EDA Report	VIDEO	 EDA Use Case.pdf	05:14
36	EDA Quiz	QUIZ	—	Assessment

SECTION 05 SQL for Data Science

19 Lectures

1 Quiz

1 Files

5h 10m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Course Contents	ARTICLE	 Resources	00:08
2	Installation	VIDEO	—	03:25
3	Data Architecture - File server vs client server	VIDEO	—	07:25
4	Introduction to SQL	VIDEO	—	09:47
5	Constraints in SQL	VIDEO	—	16:16
6	Table Basics - DDLs	VIDEO	—	22:52
7	Table Basics - DQLs	VIDEO	—	17:10
8	Table Basics - DMLs	VIDEO	—	28:20
9	Joins	VIDEO	—	25:59
10	Data Import Export	VIDEO	—	31:47
11	Aggregation Functions	VIDEO	—	12:01
12	String functions	VIDEO	—	16:47
13	Date Time Functions	VIDEO	—	13:34
14	Regular Expressions	VIDEO	—	09:18
15	Nested Queries	VIDEO	—	14:57
16	Views	VIDEO	—	12:43
17	Stored Procedures	VIDEO	—	25:33
18	Windows Function	VIDEO	—	22:12
19	SQL Python connectivity	VIDEO	—	20:08
20	SQL Quiz	QUIZ	—	Assessment

SECTION 06 Machine Learning

86 Lectures

1 Quiz

1 Files

20h 13m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Agenda	VIDEO	 Resources	02:24
2	Introduction to ML	VIDEO	—	04:51
3	Types of ML	VIDEO	—	18:17
4	Use Cases Part 1	VIDEO	—	02:10
5	Use Cases Part 2	VIDEO	—	01:28
6	Pre-Requisites: Features	VIDEO	—	11:37
7	Pre-Requisites: Train-Test Split	VIDEO	—	18:04
8	Pre-Requisites: Feature Scaling	VIDEO	—	11:04
9	Pre-Requisites: Standardization Example	VIDEO	—	04:13
10	Pre-Requisites: Normalization Example	VIDEO	—	02:30
11	Pre-Requisites: Feature Encoding	VIDEO	—	12:25
12	Pre-Requisites: Feature Encoding (Practicals)	VIDEO	—	10:46
13	Regression: Introduction to Regression Models	VIDEO	—	05:44
14	Regression: Regression Metrics	VIDEO	—	35:20
15	Regression: Regression Metrics (Practicals)	VIDEO	—	18:18
16	Regression: Simple Linear Regression	VIDEO	—	15:44
17	Regression: Multiple Linear Regression	VIDEO	—	11:33
18	Regression: Linear Regression (Practicals)	VIDEO	—	26:13
19	Regression: Multiple Linear Regression (Practicals)	VIDEO	—	13:24
20	Regression: Polynomial Regression	VIDEO	—	10:10
21	Regression: Polynomial Regression (Practicals)	VIDEO	—	26:30
22	Regression: Bias Variance Tradeoff	VIDEO	—	09:15
23	Regression: Ridge Regression	VIDEO	—	13:31
24	Regression: Lasso Regression	VIDEO	—	09:06
25	Regression: Lasso & Ridge Regression (Practicals)	VIDEO	—	44:53
26	Classification: Introduction to Classification	VIDEO	—	06:04

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
27	Classification: Types of Classification	VIDEO	—	06:39
28	Classification: Log Loss	VIDEO	—	15:53
29	Classification: Confusion Matrix	VIDEO	—	13:00
30	Classification: AUC ROC Curve	VIDEO	—	09:14
31	Classification: Classification Report	VIDEO	—	07:39
32	Classification: kNN Classifier	VIDEO	—	11:01
33	Classification: kNN Classifier Example	VIDEO	—	10:20
34	Classification: Practicals Part 1	VIDEO	—	14:31
35	Classification: kNN Classifier (Practicals)	VIDEO	—	14:16
36	Classification: Decision Tree	VIDEO	—	08:37
37	Classification: Decision Tree (Entropy based)	VIDEO	—	14:17
38	Classification: Decision Tree (gini based)	VIDEO	—	20:08
39	Classification: Decision Tree (Practicals)	VIDEO	—	08:02
40	Classification: Decision Tree (Visualizing)	VIDEO	—	21:04
41	Classification: Random Forest Classifier	VIDEO	—	05:56
42	Classification: Random Forest Classifier (Practicals)	VIDEO	—	05:05
43	Classification: Naive Bayes Classifier	VIDEO	—	14:45
44	Classification: SVM Classifier Part 1	VIDEO	—	14:59
45	Classification: SVM Classifier Part 2	VIDEO	—	14:29
46	Classification: Logistic Regression	VIDEO	—	19:02
47	Classification: Practicals so far	VIDEO	—	27:48
48	Classification: Issues in Classification (Part 1)	VIDEO	—	08:21
49	Classification: Issues in Classification (Part 2)	VIDEO	—	11:00
50	Classification: Project	VIDEO	—	35:14
51	Ensemble: Introduction to Ensemble Learning	VIDEO	—	28:31
52	Ensemble: Bagging	VIDEO	—	15:55
53	Ensemble: Bagging vs Random Forest	VIDEO	—	13:23
54	Ensemble: Bagging (Practicals #1)	VIDEO	—	31:09

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
55	Ensemble: Bagging (Practicals #2)	VIDEO	—	20:31
56	Ensemble: Boosting	VIDEO	—	11:01
57	Ensemble: Ada Boost	VIDEO	—	23:18
58	Ensemble: Gradient Boost	VIDEO	—	06:40
59	Ensemble: CF vs LF	VIDEO	—	10:12
60	Ensemble: Cross Entropy	VIDEO	—	06:47
61	Ensemble: Xtreme Gradient Boosting (XGB)	VIDEO	—	18:36
62	Ensemble: Project	VIDEO	—	24:51
63	Clustering: Introduction to Clustering	VIDEO	—	16:37
64	Clustering: kMeans Clustering	VIDEO	—	20:29
65	Clustering: kMeans Clustering (Practicals)	VIDEO	—	19:53
66	Clustering: Hierarchical Clustering	VIDEO	—	15:48
67	Clustering: Hierarchical Clustering (Practicals)	VIDEO	—	14:02
68	Clustering: Mean Shift Clustering	VIDEO	—	11:28
69	Feature Engineering: Introduction	VIDEO	—	13:15
70	Feature Engineering: RFE and SFS	VIDEO	—	06:39
71	Feature Engineering: RFE (Practicals)	VIDEO	—	23:18
72	Feature Engineering: Successive Feature Selection	VIDEO	—	19:30
73	Feature Engineering: Chi-Square	VIDEO	—	06:03
74	Feature Engineering: Chi-Square (Practicals)	VIDEO	—	06:19
75	Feature Engineering: Principal Component Analysis	VIDEO	—	34:47
76	Feature Engineering: Principal Component Analysis (Practicals)	VIDEO	—	10:26
77	Feature Engineering: Linear Discriminant Analysis	VIDEO	—	09:52
78	Feature Engineering: Linear Discriminant Analysis (Practicals)	VIDEO	—	09:31
79	Feature Engineering: kPCA & QDA	VIDEO	—	06:14
80	Feature Engineering: kPCA & QDA (Practicals)	VIDEO	—	05:41
81	Hyper Parameter Optimization: Basics	VIDEO	—	10:05

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
82	Hyper Parameter Optimization: Manual HPO	VIDEO	—	06:07
83	Hyper Parameter Optimization: GridSearch vs RandomizedSearch	VIDEO	—	10:58
84	Hyper Parameter Optimization: Manual HPO (Practicals)	VIDEO	—	22:12
85	Hyper Parameter Optimization: RandomizedSearchCV (Practicals)	VIDEO	—	18:18
86	Hyper Parameter Optimization: GridSearchCV (Practicals)	VIDEO	—	08:20
87	ML Quiz	QUIZ	—	Assessment

SECTION 07 Time Series Analysis & Forecasting

60 Lectures

1 Quiz

1 Files

7h 29m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction to TSA	VIDEO	 Resources	04:29
2	Time Series vs Regression	VIDEO	—	10:40
3	Time Series Analysis	VIDEO	—	02:27
4	Anomaly Detection	VIDEO	—	04:36
5	Components of Time Series	VIDEO	—	09:09
6	Decomposition	VIDEO	—	01:54
7	Decomposition (Practicals)	VIDEO	—	06:36
8	Additive/Multiplicative Decomposition	VIDEO	—	08:41
9	Stationarity	VIDEO	—	08:54
10	Testing TS Stationarity	VIDEO	—	07:03
11	Transformation	VIDEO	—	06:10
12	Introduction to Pre-Processing	VIDEO	—	03:36
13	Handle Missing Value	VIDEO	—	06:25
14	Handle Missing Value (Practicals)	VIDEO	—	11:35
15	Outlier Treatment	VIDEO	—	09:21
16	3-Sigma Technique	VIDEO	—	13:41
17	Feature Scaling	VIDEO	—	11:04
18	Feature Scaling: Standardization	VIDEO	—	04:13
19	Feature Scaling: Normalization	VIDEO	—	02:30
20	Feature Scaling (Practicals)	VIDEO	—	13:50
21	Feature Encoding	VIDEO	—	12:25
22	Feature Encoding (Practicals)	VIDEO	—	10:46
23	Models - Algorithms	VIDEO	—	01:47
24	Models - ARIMA Part 1	VIDEO	—	03:45
25	Models - ARIMA Part 2	VIDEO	—	07:26
26	Models - AR Theory	VIDEO	—	08:37

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
27	Models - MA Theory	VIDEO	—	09:40
28	Models - ACF/PACF Plots	VIDEO	—	13:37
29	Models - Find p,d,q in ARIMA	VIDEO	—	03:33
30	Models - ARIMA (Practicals Part 1)	VIDEO	—	14:12
31	Models - ARIMA (Practicals Part 2)	VIDEO	—	12:53
32	Models - ARIMA (Final)	VIDEO	—	10:06
33	Models - Decomposition	VIDEO	—	03:26
34	Models - ACF/PACF	VIDEO	—	03:38
35	Models - Best Transformation	VIDEO	—	10:16
36	Models - Grid Search (Part 1)	VIDEO	—	08:49
37	Models - Grid Search (Part 2)	VIDEO	—	02:04
38	Models - Final Model Building	VIDEO	—	11:04
39	Models - Facebook Prophet (Part 1)	VIDEO	—	07:56
40	Models - Facebook Prophet (Part 2)	VIDEO	—	11:05
41	Models - Facebook Prophet (Part 3)	VIDEO	—	06:07
42	Models - Multi Variate Time Series Analysis	VIDEO	—	06:35
43	Models - Facebook Prophet (Uni vs Multi)	VIDEO	—	13:33
44	Introduction to Metrics	VIDEO	—	05:12
45	Forecasting Evaluation Metrics	VIDEO	—	02:49
46	Mean Squarred Error	VIDEO	—	02:45
47	Root Mean Squarred Error	VIDEO	—	02:12
48	Mean Absolute Percentage Error	VIDEO	—	04:47
49	Project 1 - Energy Forecasting Part 1	VIDEO	—	03:10
50	Project 1 - Energy Forecasting Part 2	VIDEO	—	06:26
51	Project 1 - Energy Forecasting Part 3	VIDEO	—	07:04
52	Project 2 - Stock Market Prediction Part 1	VIDEO	—	04:42
53	Project 2 - Stock Market Prediction Part 2	VIDEO	—	04:07
54	Project 2 - Stock Market Prediction Part 3	VIDEO	—	13:48

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
55	Project 3 - Demand Forecasting Part 1	VIDEO	—	03:32
56	Project 3 - Demand Forecasting Part 2	VIDEO	—	13:32
57	Project 3 - Demand Forecasting Part 3	VIDEO	—	09:57
58	Project 3 - Demand Forecasting Part 4	VIDEO	—	01:30
59	Project 3 - Demand Forecasting Part 5	VIDEO	—	15:14
60	Project 3 - Demand Forecasting Part 6	VIDEO	—	08:34
61	TSA Quiz	QUIZ	—	Assessment

SECTION 08 Deep Learning & Neural Networks

34 Lectures

1 Quiz

1 Files

5h 57m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction	ARTICLE	 Resources	01:51
2	Introduction to Deep Learning	VIDEO	—	03:02
3	Understanding Deep Learning	VIDEO	—	05:58
4	What is a Neuron	VIDEO	—	08:48
5	Activation Functions	VIDEO	—	05:12
6	Activation Function: Step Function	VIDEO	—	06:48
7	Activation Function: Linear Function	VIDEO	—	13:32
8	Activation Function: Sigmoid Function	VIDEO	—	07:06
9	Activation Function: TanH Function	VIDEO	—	03:35
10	Activation Function: ReLu Function	VIDEO	—	10:18
11	Backpropagation & Forward Pass	VIDEO	—	15:42
12	Gradient Descent	VIDEO	—	07:32
13	Artificial Neural Networks: Intuition	VIDEO	—	04:33
14	Artificial Neural Networks: Practicals	VIDEO	—	15:25
15	Artificial Neural Networks: Hyper Parameter Optimization	VIDEO	—	12:37
16	Convolutional Neural Networks: What is CNN	VIDEO	—	08:07
17	Convolutional Neural Networks: Steps in CNN	VIDEO	—	12:19
18	Convolutional Neural Networks: Architecture Explained	VIDEO	—	17:21
19	Convolutional Neural Networks: Image Augmentation	VIDEO	—	14:40
20	Convolutional Neural Networks: Batch size vs iterations vs epochs	VIDEO	—	09:19
21	Convolutional Neural Networks: Practicals	VIDEO	—	21:21
22	Convolutional Neural Networks: Model Summary & Parameters	VIDEO	—	08:14
23	Convolutional Neural Networks: Project (X-Ray detection)	VIDEO	—	17:53
24	Recurrent Neural Networks: Basics	VIDEO	—	06:52

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
25	Recurrent Neural Networks: Types of RNN	VIDEO	—	05:18
26	Recurrent Neural Networks: Vanishing Gradient & Exploding Gradient Problem	VIDEO	—	12:53
27	Recurrent Neural Networks: LSTMs	VIDEO	—	05:55
28	Recurrent Neural Networks: LSTMs (Practicals)	VIDEO	—	11:13
29	Pre-Trained Models	VIDEO	—	22:37
30	Pre-Trained Models (Practicals)	VIDEO	—	23:23
31	Pre-Trained Models: VGG16	VIDEO	—	09:47
32	Pre-Trained Models: MobileNet	VIDEO	—	05:40
33	Transfer Learning	VIDEO	—	07:30
34	Project: Pneumonia Detection from X-Ray Images	VIDEO	—	15:13
35	Deep learning Quiz	QUIZ	—	Assessment

SECTION 09 Natural Language Processing

47 Lectures

1 Quiz

3 Files

9h 21m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Course Contents	ARTICLE	 Resources	01:33
2	Intro to NLP: Introduction	VIDEO	—	06:57
3	Intro to NLP: Introduction continued	VIDEO	—	06:08
4	Intro to NLP: Key Challenges	VIDEO	—	09:02
5	Intro to NLP: Linguistics	VIDEO	—	07:08
6	NLP Basics: Case Folding	VIDEO	—	04:53
7	NLP Basics: SCR	VIDEO	—	14:17
8	NLP Basics: Handling Contractions	VIDEO	—	10:01
9	NLP Basics: Tokenization	VIDEO	—	05:55
10	NLP Basics: Stop Word Removal	VIDEO	—	07:09
11	NLP Basics: nGrams	VIDEO	—	08:34
12	NLP Basics: Vectorization	VIDEO	—	04:47
13	NLP Basics: Word Embeddings	VIDEO	—	02:37
14	NLP Basics: Bag of Words	VIDEO	—	09:48
15	NLP Basics: Bag of Words (Practicals)	VIDEO	—	19:00
16	NLP Basics: TF-IDF	VIDEO	—	12:18
17	NLP Basics: TF-IDF (Practicals)	VIDEO	—	21:19
18	NLP Basics: Part of Speech Tagging and Named Entity Recognition	VIDEO	—	12:42
19	NLP Basics: NER (Practicals)	VIDEO	—	12:46
20	Word Embeddings: Word2Vec Introduction	VIDEO	—	06:18
21	Word Embeddings: Word2Vec Part 2	VIDEO	—	03:24
22	Word Embeddings: Pre-Trained Word2Vec	VIDEO	—	07:34
23	Word Embeddings: Word2Vec Intuition	VIDEO	—	08:50
24	Word Embeddings: Word2Vec - Check X Features	VIDEO	—	10:01
25	Word Embeddings: Word2Vec CBOW	VIDEO	—	19:10
26	Word Embeddings: Word2Vec Skip Grams	VIDEO	—	11:23

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
27	Word Embeddings: GloVe	VIDEO	—	18:04
28	Word Embeddings: FastText	VIDEO	—	11:58
29	GloVe vs FastText (Practicals #1)	VIDEO	 GloVe_vs_FastText_(Practical)  wordsim353.tsv	10:16
30	GloVe vs FastText (Practicals #2)	VIDEO	—	16:22
31	Word Embeddings: Cosine Similarity	VIDEO	—	18:22
32	Neural Networks: LSTMs Part 1	VIDEO	—	19:25
33	Neural Networks: LSTMs Part 2 (Architecture)	VIDEO	—	16:19
34	Neural Networks: LSTMs Part 3 (Deep Dive)	VIDEO	—	06:53
35	Neural Networks: LSTMs Part 4 (Pointwise Operations)	VIDEO	—	06:19
36	Neural Networks: LSTMs Part 5 (forget gate)	VIDEO	—	12:45
37	Neural Networks: LSTMs Part 6 (inpute gate)	VIDEO	—	18:29
38	Neural Networks: LSTMs Part 7 (output gate)	VIDEO	—	09:38
39	Neural Networks: LSTMs Part 8 (Practicals #1)	VIDEO	—	33:39
40	Neural Networks: LSTMs Part 9 (Practicals #2)	VIDEO	—	17:20
41	Neural Networks: LSTMs Part 10 (Practicals #3)	VIDEO	—	13:22
42	Neural Networks: GRU Part 1	VIDEO	—	05:16
43	Neural Networks: GRU Part 2	VIDEO	—	25:48
44	Neural Networks: GRU Part 3 (reset gate)	VIDEO	—	08:39
45	Neural Networks: GRU Part 4 (update gate)	VIDEO	—	09:47
46	Neural Networks: GRU Part 5 (Practicals)	VIDEO	—	12:07
47	Neural Networks: Bi-Directional LSTMs	VIDEO	—	27:14
48	NLP Quiz	QUIZ	—	Assessment

SECTION 10 Transformers & Generative AI

53 Lectures

1 Quiz

9 Files

11h 12m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Course Contents	ARTICLE	—	01:49
2	Transformer Types	VIDEO	—	19:43
3	Introduction to Transformers	VIDEO	—	33:08
4	Self Attention	VIDEO	—	27:25
5	Encoder Architecture	VIDEO	—	09:10
6	Contextual Embeddings	VIDEO	—	05:39
7	Decoder Architecture	VIDEO	—	05:29
8	Introduction to BERT	VIDEO	—	13:33
9	Configurations of BERT	VIDEO	—	03:46
10	BERT: Fine Tuning	VIDEO	—	03:48
11	BERT: Pre Tuning (Masked LM)	VIDEO	—	08:37
12	BERT: Input Embeddings	VIDEO	—	09:13
13	ARLM vs AELM	VIDEO	—	06:23
14	RoBERTa	VIDEO	—	09:26
15	DistilBERT	VIDEO	—	13:48
16	AlBERT	VIDEO	—	14:20
17	Introduction to GPT (Decoder Only)	VIDEO	—	05:44
18	GPT Architecture	VIDEO	—	05:21
19	GPT Masked Multi Head Attention	VIDEO	—	11:49
20	GPT Blocks	VIDEO	—	07:21
21	GPT Training	VIDEO	—	09:37
22	LLM Basics: Tokens	ARTICLE	—	01:53
23	LLM Basics: Context Window	VIDEO	—	10:58
24	LLM Basics: Prompt	VIDEO	—	10:50
25	LLM Basics: Prompt Engineering	VIDEO	—	18:25
26	LLM Basics: Prompt Tuning	VIDEO	—	06:58
27	LLM Basics: Prompt Structures	VIDEO	—	13:25

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
28	RAGs: Introduction to RAG	VIDEO	—	01:36
29	RAGs: What and Why	VIDEO	—	19:03
30	RAGs: Use Cases	VIDEO	—	16:45
31	RAGs: Paper Explanation	VIDEO	—	05:43
32	RAGs: Architecture Explanation	VIDEO	—	19:07
33	RAGs: Detailed Architecture Walkthrough	VIDEO	—	12:44
34	RAGs: Practical Use Cases	VIDEO	 custom_pdf_chatbot.ipynb  RAGPaper (1).pdf	33:56
35	LangChain	VIDEO	—	10:31
36	Introduction to Prompt Engineering	VIDEO	—	11:05
37	Types of Prompting	VIDEO	—	13:51
38	Few Shot Limitations	VIDEO	—	07:18
39	Chain of Thoughts Prompting	VIDEO	—	06:54
40	Vector Databases	VIDEO	—	13:00
41	Vector Database vs Vector Index	VIDEO	—	08:04
42	How Vector Databases works	VIDEO	—	10:43
43	Vector Database (Practicals)	VIDEO	 custom_pdf_chatbot(Pinecone  RAGPaper (1).pdf	30:41
44	LSH	VIDEO	—	14:29
45	Model Overview: Ollama	VIDEO	—	17:44
46	Getting Started: Ollama	VIDEO	—	18:45
47	Model Testing: Ollama	VIDEO	—	22:25
48	Python Implementation: Ollama	VIDEO	—	07:51
49	RAG Systems: Ollama	VIDEO	—	13:25
50	RAG Systems (Practicals): Ollama	VIDEO	 RAG with Ollama.ipynb  RAGPaper.pdf	25:55
51	Model Overview: LLM APIs	VIDEO	—	26:31
52	RAG Systems with xAI: LLM APIs	VIDEO	—	04:54
53	RAG Systems with xAI (Practicals): LLM APIs	VIDEO	 RAG with GrokAI.ipynb  Power BI Ebook.pdf	11:56

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
			 RAGPaper.pdf	
54	Gen AI Quiz	QUIZ	—	Assessment

SECTION 11 Generative AI (Advanced) 9 Lectures 1 Quiz 1h 47m				
#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction to Fine Tuning	VIDEO	—	11:12
2	RAGs vs Fine-Tuning	VIDEO	—	06:25
3	When to use RAGs vs When to use Fine-Tuned models	VIDEO	—	11:28
4	Fine Tuning	VIDEO	—	11:46
5	PEFT and Quantization	VIDEO	—	17:48
6	LoRA (Phase I - Low Rank)	VIDEO	—	19:40
7	LoRA (Phase II - Adapters)	VIDEO	—	11:16
8	LoRA (Phase III - Low Rank + Adapters)	VIDEO	—	10:10
9	qLoRA (Quantized Low Rank Adapters)	VIDEO	—	08:10
10	Advanced Gen AI Quiz	QUIZ	—	Assessment

SECTION 12 RAG Assessment and Evaluation Metrics

12 Lectures

1 Quiz

3 Files

3h 45m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction	ARTICLE	—	01:05
2	Introduction to RAG Evaluation	VIDEO	—	07:54
3	N-Gram Metrics (BLEU)	VIDEO	—	14:08
4	N-Gram Metrics (ROUGE)	VIDEO	—	07:39
5	N-Gram Metrics (METEOR)	VIDEO	—	04:17
6	Intrinsic Metrics (PPL Score)	VIDEO	—	21:57
7	Intrinsic Metrics (PPL Score) Practicals	VIDEO	 RAG_Metrics_PPL_Score.ipynb	11:21
8	Model-based Metrics (BARTScore vs BERTScore)	VIDEO	—	13:36
9	RAGAS Framework (RAG Assessment)	VIDEO	—	58:57
10	(PRACTICALS) Rag Evaluation by RAGAS	VIDEO	 Power BI Ebook.pdf  RAG_Metrics_Evaluation.ipynb	44:24
11	(PRACTICALS) Rag Evaluation by BLEU, ROUGE	VIDEO	—	13:28
12	(PRACTICALS) Rag Evaluation by BERTScore, BARTScore	VIDEO	—	26:37
13	RAG Evaluation Quiz	QUIZ	—	Assessment

SECTION 13 Introduction to Agentic AI

19 Lectures

5 Files

3h 24m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Download the Course Resources	ARTICLE	📄 Agent.ipynb 📄 Tool_Calling.ipynb 📄 Tool_Calling_(Agentic_AI).ip 📄 Tools_(Agentic_AI).ipynb 📄 Currency_Conversion_Tool.ipy	00:05
2	Introduction	VIDEO	—	18:00
3	Agentic AI - Examples	VIDEO	—	09:19
4	What are Tools?	VIDEO	—	06:56
5	LangChain (Built-in tools)	VIDEO	—	11:16
6	Custom Tools using Tool Decorator	VIDEO	—	12:30
7	Custom Tools using StructuredTool	VIDEO	—	04:03
8	Custom Tools using BaseTool	VIDEO	—	04:33
9	LangChain Toolkits	VIDEO	—	10:32
10	LangChain Tool Calling Part #1	VIDEO	—	02:49
11	LangChain Tool Calling Part #2	VIDEO	—	13:16
12	Tool Execution	VIDEO	—	10:07
13	Currency Conversion Tool (Mini Project)	VIDEO	—	41:15
14	Travel Planning Agent Part #1	VIDEO	—	14:50
15	Travel Planning Agent Part #2	VIDEO	—	11:08
16	Travel Planning Agent Part #3	VIDEO	—	06:01
17	AI Agent (Practicals)	VIDEO	—	13:12
18	AI Agent Code Explanation and Theory Part #1	VIDEO	—	07:53
19	AI Agent Code Explanation and Theory Part #2	VIDEO	—	06:29

SECTION 14 Agentic AI using LangGraph

13 Lectures

1 Files

1h 33m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction	VIDEO	—	02:15
2	Gen AI vs Traditional AI	VIDEO	—	05:16
3	Gen AI vs Agentic AI	VIDEO	—	07:02
4	Scenario #1	VIDEO	—	07:37
5	Key Characteristics of an Agentic AI workflow	VIDEO	—	01:38
6	Autonomy	VIDEO	—	06:15
7	Goal Oriented	VIDEO	—	02:27
8	Planning	VIDEO	—	03:47
9	Reasoning	VIDEO	—	04:11
10	Adaptability	VIDEO	—	02:38
11	Context Awareness	VIDEO	—	03:38
12	Components of an Agentic AI Workflow	VIDEO	—	07:05
13	Mini Project using LangGraph	VIDEO	 Agentic AI (Mini Project).ip	39:23

SECTION 15 No-Code Tools for Agentic AI

9 Lectures

3 Files

2h 02m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction to n8n	VIDEO	—	05:04
2	n8n - Templates	VIDEO	—	01:41
3	n8n - Tool Demo	VIDEO	—	07:45
4	n8n - Project Overview	VIDEO	📄 My workflow.json 📄 Student_engagement (New).xls	08:00
5	n8n - Workflow Overview	VIDEO	—	26:30
6	n8n - Workflow from Scratch	VIDEO	—	41:13
7	n8n - AI Powered News Summarizer (Part 1)	VIDEO	📄 AI Powered News Summarizer P	13:08
8	n8n - AI Powered News Summarizer (Part 2)	VIDEO	—	17:12
9	n8n - AI Powered News Summarizer (Part 3)	VIDEO	—	01:30

SECTION 16 ML/DL Deployment

7 Lectures

1 Files

1h 49m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Deployment Basics	VIDEO	📄 Resources	07:27
2	Introduction to Flask	VIDEO	—	12:41
3	Flask Basic App	VIDEO	—	12:37
4	Model Building (Breast Cancer Prediction)	VIDEO	—	15:55
5	Flask App (Breast Cancer Prediction)	VIDEO	—	22:23
6	AWS	VIDEO	—	09:52
7	AWS Deployment (Breast Cancer Prediction)	VIDEO	—	28:51

SECTION 17 Data Engineering Basics

11 Lectures

1 Quiz

39m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction to Data Engineering	VIDEO	—	00:39
2	What is ETL	VIDEO	—	07:48
3	ETL Tools	VIDEO	—	05:55
4	What is Data Warehouse	VIDEO	—	03:13
5	Benefits of Data Warehouse	VIDEO	—	01:29
6	Data Warehouse Structure	VIDEO	—	03:06
7	Why do we need Staging	VIDEO	—	06:36
8	What are Data Marts	VIDEO	—	02:56
9	Data Lake	VIDEO	—	02:32
10	Data lake vs Data Warehouse	VIDEO	—	03:44
11	Elements of Datalake	VIDEO	—	01:50
12	Data Engineering Quiz	QUIZ	—	Assessment

SECTION 18 Machine Learning Project #1 - Cancer Risk Prediction

22 Lectures

3 Files

2h 53m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Problem Statement	VIDEO	 Cancer Prediction - Report.p  Deployment.txt  Cancer_Risk_prediction - Fin	02:53
2	Problem Statement (Technical)	VIDEO	—	01:44
3	Architecture Diagram	VIDEO	—	07:17
4	Data Understanding	VIDEO	—	10:53
5	EDA #1	VIDEO	—	11:36
6	EDA #2	VIDEO	—	05:35
7	EDA #3	VIDEO	—	03:32
8	EDA #4	VIDEO	—	15:18
9	EDA #5	VIDEO	—	12:40
10	EDA #6	VIDEO	—	05:44
11	Model Building #1	VIDEO	—	22:04
12	Model Building #2	VIDEO	—	05:17
13	Model Building #3	VIDEO	—	05:57
14	Model Building #4	VIDEO	—	11:08
15	Model Building #5	VIDEO	—	04:46
16	Model Building #6 (XgBoost)	VIDEO	—	06:56
17	Model Building #7 (XgBoost + Optuna)	VIDEO	—	06:37
18	Model Saving	VIDEO	—	02:12
19	Streamlit Application	VIDEO	—	08:45
20	Deployment #1	VIDEO	—	02:43
21	Deployment #2	VIDEO	—	02:01
22	Deployment #3	VIDEO	—	18:00

SECTION
19
Deep Learning Project #1 - Solar Panel Defect Classification

 18
 Lectures

 2
 Files

 2h
 34m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction the Project	VIDEO	📁 Data.zip 📁 Code.zip	05:04
2	Data Understanding	VIDEO	—	05:02
3	Architecture Diagram	VIDEO	—	10:29
4	Running the base CNN2d model	VIDEO	—	17:43
5	Base CNN2d model (10 epochs)	VIDEO	—	04:32
6	Base CNN2d model (20 epochs)	VIDEO	—	01:34
7	Hit & Trial #1	VIDEO	—	04:48
8	Hit & Trial #2	VIDEO	—	06:49
9	Hit & Trial #3	VIDEO	—	00:57
10	Transfer Learning - MobileNetV2	VIDEO	—	08:40
11	Transfer Learning - MobileNetV2 (20 epochs)	VIDEO	—	02:23
12	Transfer Learning - MobileNetV2 (20 epochs - output)	VIDEO	—	01:49
13	Transfer Learning - EfficientNetB0	VIDEO	—	15:57
14	Transfer Learning - EfficientNetB0 - Hyper Parameter Optimization	VIDEO	—	10:03
15	Transfer Learning - EfficientNetB0 - HPO Results	VIDEO	—	01:58
16	Convert to Streamlit App	VIDEO	—	25:10
17	Streamlit Code Walkthrough	VIDEO	—	08:54
18	Deployment on AWS EC2	VIDEO	—	22:14

**SECTION
20**
**Deep Learning Project #2 - Elephant Species
Classification**

 16
Lectures

 3
Files

 2h
38m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction	VIDEO	 test-20260105T160050Z-1-001.  train-20260105T160050Z-1-001  Code.zip	07:03
2	Business Problem	VIDEO	—	06:20
3	Architecture Diagram	VIDEO	—	09:07
4	Data Understanding	VIDEO	—	04:34
5	Base Model (10 Epochs)	VIDEO	—	27:15
6	Base Model Building	VIDEO	—	03:54
7	Base Model (100 Epochs)	VIDEO	—	09:51
8	Transfer Learning (Theory)	VIDEO	—	05:44
9	MobileNetV2 - Part 1	VIDEO	—	10:00
10	MobileNetV2 - Part 2	VIDEO	—	03:02
11	MobileNetV2 - Part 3	VIDEO	—	07:12
12	MobileNetV2 - Part 4	VIDEO	—	06:04
13	Xception - Part 1	VIDEO	—	09:10
14	Xception - Part 2	VIDEO	—	01:29
15	Streamlit Application	VIDEO	—	23:31
16	AWS EC2 Deployment	VIDEO	—	24:00

SECTION 21 Generative AI Projects

32 Lectures

10 Files

8h 56m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Project 1: ChatScholar (EdTech Project)	VIDEO	GitHub	46:28
2	Project 2: Research RAG Chatbot	VIDEO	GitHub	35:29
3	Project 3: Automated AI Claims Processing using Gen AI	VIDEO	GitHub	47:27
4	Project 4: Multi PDF RAG Chatbot built on Web Scraped Data	VIDEO	GitHub	34:55
5	Project 5: AI Career Coach: Part 1	VIDEO	FREE PREVIEW Google Drive	13:37
6	Project 5: AI Career Coach: Part 2	VIDEO	—	11:35
7	Project 5: AI Career Coach: Part 3	VIDEO	—	24:56
8	Project 6: Sustainability Chatbot (using xAI's Grok-3 AI)	VIDEO	Drive	47:38
9	Project 7: Text-to-SQL Chatbot: Part 1	VIDEO	Code Dataset	11:33
10	Project 7: Text-to-SQL Chatbot: Part 2	VIDEO	—	21:38
11	Project 7: Text-to-SQL Chatbot: Part 3	VIDEO	—	13:08
12	Project 7: Text-to-SQL Chatbot: Part 4	VIDEO	—	19:59
13	Project 7: Text-to-SQL Chatbot: Part 5	VIDEO	—	14:14
14	Project 7: Text-to-SQL Chatbot: Part 6	VIDEO	—	10:17
15	Project 7: Text-to-SQL Chatbot: Part 7	VIDEO	—	05:42
16	Project 8: RAG Chatbot (using AWS Bedrock): Part 1	VIDEO	Air India RAG Chatbot (Codeb	07:28
17	Project 8: RAG Chatbot (using AWS Bedrock): Part 2	VIDEO	—	05:29
18	Project 8: RAG Chatbot (using AWS Bedrock): Part 3	VIDEO	—	03:31
19	Project 8: RAG Chatbot (using AWS Bedrock): Part 4	VIDEO	—	06:58
20	Project 8: RAG Chatbot (using AWS Bedrock): Part 5	VIDEO	—	02:50
21	Project 8: RAG Chatbot (using AWS Bedrock): Part 6	VIDEO	—	06:53
22	Project 8: RAG Chatbot (using AWS Bedrock): Part 7	VIDEO	—	15:42
23	Project 9: Resume Genie (A Gen AI Powered Resume Platform)	ARTICLE	resume genie.zip	00:51
24	Project 9: Introduction	VIDEO	—	04:27

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
25	Project 9: Overview	VIDEO	—	04:53
26	Project 9: Getting started with Streamlit	VIDEO	—	09:33
27	Project 9: Resume Checker	VIDEO	—	17:56
28	Project 9: Cover Letter Generator	VIDEO	—	14:00
29	Project 9: Resume Scorer	VIDEO	—	16:16
30	Project 9: AI Career Coach	VIDEO	—	20:53
31	Project 9: Streamlit App (Final)	VIDEO	—	13:25
32	Project 9: AWS Deployment	VIDEO	—	27:04

SECTION 22 Agentic AI Projects

7 Lectures

1 Files

1h 24m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Dental Appointment System (Part 1)	VIDEO	 GitHub	10:10
2	Dental Appointment System (Part 2)	VIDEO	—	12:59
3	Dental Appointment System (Part 3)	VIDEO	—	05:49
4	Dental Appointment System (Part 4)	VIDEO	—	15:24
5	Dental Appointment System (Part 5)	VIDEO	—	19:26
6	Dental Appointment System (Part 6)	VIDEO	—	10:31
7	Dental Appointment System (Part 7)	VIDEO	—	10:16

SECTION 23 Generative AI Deployment

9 Lectures

1 Files

58m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Introduction to Model Deployment	VIDEO	 Code	08:23
2	Architecture Discussion	VIDEO	—	02:40
3	Model Serving	VIDEO	—	04:53
4	Next Steps	VIDEO	—	02:24
5	Air India Chatbot	VIDEO	—	02:21
6	Coding - PDF Based	VIDEO	—	12:11
7	Coding - Directory Based	VIDEO	—	04:38
8	Streamlit Application (Vibe Coding)	VIDEO	—	04:53
9	AWS EC2 Deployment	VIDEO	—	16:29

SECTION 24 Interview Prep (NEW SECTION)

31 Lectures

1 Files

4h 51m

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	ML Interview Prep	VIDEO	 Download Interview QnA	02:15
2	ML Interview #1	VIDEO	—	11:45
3	ML Interview #2	VIDEO	—	15:04
4	ML Interview #3	VIDEO	—	10:13
5	ML Interview #4	VIDEO	—	10:06
6	ML Interview #5	VIDEO	—	06:38
7	ML Interview #6	VIDEO	—	10:33
8	ML Interview #7	VIDEO	—	08:39
9	ML Interview #8	VIDEO	—	09:59
10	ML Interview #9	VIDEO	—	13:18
11	ML Interview #10	VIDEO	—	09:53
12	DL Interview #1	VIDEO	—	11:31
13	DL Interview #2	VIDEO	—	07:38
14	DL Interview #3	VIDEO	—	12:53
15	DL Interview #4	VIDEO	—	07:15
16	DL Interview #5	VIDEO	—	08:11
17	DL Interview #6	VIDEO	—	10:16
18	DL Interview #7	VIDEO	—	11:44
19	DL Interview #8	VIDEO	—	09:51
20	DL Interview #9	VIDEO	—	04:03
21	DL Interview #10	VIDEO	—	09:01
22	Gen AI Interview #1	VIDEO	—	05:36
23	Gen AI Interview #2	VIDEO	—	06:03
24	Gen AI Interview #3	VIDEO	—	11:24
25	Gen AI Interview #4	VIDEO	—	08:10
26	Gen AI Interview #5	VIDEO	—	08:29

#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
27	Gen AI Interview #6	VIDEO	—	10:00
28	Gen AI Interview #7	VIDEO	FREE PREVIEW	07:58
29	Gen AI Interview #8	VIDEO	—	10:51
30	Gen AI Interview #9	VIDEO	—	10:32
31	Gen AI Interview #10	VIDEO	—	11:50

SECTION 25 Real-Time Client Discussions (Role Play) 0 Lectures 4 Role-Plays 0m				
#	CURRICULUM ITEM TITLE	FORMAT	FEATURES / RESOURCES	DURATION
1	Building an internal RAG chatbot for Capstone Pvt Ltd.	ROLE PLAY	—	Interactive
2	Building a Secure Customer-Churn Classification Model	ROLE PLAY	—	Interactive
3	Designing a RAG-Powered Website Chatbot	ROLE PLAY	—	Interactive
4	Demand Forecasting for Inventory Optimization	ROLE PLAY	—	Interactive

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